Alex Massucco (He/Him)

St. John's College, Cambridge, UK
am3270@cam.ac.uk ⋄ (+39) 366-4203514
LinkedIn

Research Interests

I am a PhD candidate in Mathematics at the University of Cambridge and St. John's College, specialising in mathematics of deep learning. My research focuses on scientific machine learning, exploring the theoretical background of neural models with a particular emphasis on understanding their structure, generalisation properties, and optimisation landscapes. I aim to develop frameworks that provide provable behavioural guarantees, contributing to the reliability and interpretability of AI systems. My broader interests lie at the intersection of pure mathematics and theoretical computer science.

Education

PhD in Applied Mathematics — October 2024 – Present
University of Cambridge - *Cambridge Image Analysis group* — *Cambridge, UK*

Supervisor: Prof. C. B. Schönlieb

Student at St. John's College — October 2024 – Present
St. John's College — *Cambridge, UK*

II Level License in Science and Technology[1] — September 2022 – July 2024
IUSS - School for Advanced Studies — *Pavia, Italy*

Thesis: *Finite Energy Non-Uniqueness of Solutions to the forced 3D Euler Equation*
Supervisor: Prof. C. De Lellis (IAS, Princeton)
Final Grade: 100/100 cum laude

MSc in Mathematics — September 2022 – July 2024
Università degli Studi di Pavia — *Pavia, Italy*

Thesis: *Existence of Weak Solutions for a Family of Fractional Diffusion Models with Aggregation*
Supervisors: Prof. S. Lisini (Università degli studi di Pavia), Prof. J. A. Carrillo (University of Oxford), Dr. A. Esposito (University of Oxford)
Final Grade: 110/110 cum laude

I Level License in Science and Technology — September 2019 – September 2022
IUSS - School for Advanced Studies — *Pavia, Italy*

Thesis: *A Preliminary Study on the Existence and Uniqueness of Minimizers for Graphon Neural Networks*
Supervisor: Prof. U. Stefanelli (University of Vienna)
Final Grade: 100/100 cum laude

BSc in Mathematics — September 2019 – September 2022
Università degli Studi di Pavia — *Pavia, Italy*

Thesis: *Regularity Issues for the Solutions of Elliptic PDEs: an overview from De Giorgi's results to today*
Supervisors: Prof. U. Gianazza (Università degli studi di Pavia), Prof. L. Ambrosio (Scuola Normale Superiore)
Final Grade: 110/110 cum laude

Alumnus of Collegio Ghislieri — September 2019 – July 2024
Collegio Ghislieri — *Pavia, Italy*

[1] Classified as a level 8 qualification in the European Qualifications Framework

Publications

Multi-Headed Transformer Architectures as Time-dependent Wasserstein Gradient Flows
arXiv preprint — *May 2026*

Muon is Not That Special: Random or Inverted Spectra Work Just as Well
arXiv preprint — *May 2026*

Deep Network Trainability via Persistent Subspace Orthogonality
arXiv preprint — *August 2025*

Finite-difference least square methods for solving Hamilton-Jacobi equations using neural networks
Journal of Computational Physics — *March 2025*

Scholarships and Awards

Cambridge Trust – CCEIT PhD Scholarship June 2024
Cambridge Trust – Three-year full scholarship covering international PhD tuition fees

CCIMI Three-Year PhD Scholarship June 2024
Cantab Capital Institute for the Mathematics of Information – Full PhD funding, including tuition fees and maintenance stipend

IUSS Support to Policymakers Scholarship December 2023
IUSS – School for Advanced Studies – Competitive academic scholarship

Erasmus+ Traineeship Scholarship June 2023
University of Pavia – Competitive mobility scholarship

Summer Research Scholarship at St John's College April 2023
Collegio Ghislieri – Fully funded one-month research scholarship, including accommodation

IUSS Premier Scholarship January 2022
IUSS – School for Advanced Studies – Competitive academic scholarship

Erasmus+ Traineeship Scholarship November 2021
University of Pavia – Competitive mobility scholarship

BSc and MSc Full Scholarship September 2019
IUSS – School for Advanced Studies – Full tuition-fee waiver and accommodation

Languages & Skills

Python	Advanced knowledge
Italian	Native Speaker
English	Professional knowledge (Certified: CAE C1)
French	Professional knowledge (Certified: DALF C1)

Research Experiences

Visiting PhD Student at Twente University April 2026
Twente University

Visiting PhD Student at Emory University February 2026 – March 2026
Emory University

Visiting Master Student at University of Oxford January 2024 – March 2024
University of Oxford

Invited Visiting Master Student at University of Texas at Austin December 2023
University of Texas at Austin

Visiting Master Student at St. John's College September 2023 – October 2023
University of Cambridge

Visiting Undergraduate Student at the University of Vienna March 2022 – May 2022
University of Vienna

Talks & Workshops

International Congress of Mathematicians 2026 — July 2026
Selected speaker (Oral) — *Philadelphia, US*

Mathematics of Imaging and AI Seminar — April 2026
Invited speaker — *Enschede, Netherlands*

Mathematics and Computer Science Talks — February 2026
Invited speaker — *Atlanta, US*

AI4X — July 2025
Selected speaker (Oral) — *Singapore, Singapore*

MIA2025 — January 2025
Selected speaker (Poster) — *Paris, France*

Optimal Control and Differential Game Workshop — December 2023
Invited speaker — *Austin, Texas*

Calculus of Variations Seminar — May 2022
Invited speaker — *Vienna, Austria*

Additional Course & Certificates

Geometric Problems in Analysis — June 2024
Scuola Galileiana — *Padova, Italy*

Project Management Open Badge — April 2024
Università degli Studi di Pavia — *Pavia, Italy*

Inverse Problems Autumn School — September 2023
Jesus College - University of Cambridge — *Cambridge, UK*

Samsung Innovation Campus — September 2022 – January 2023
Università degli Studi di Pavia — *Pavia, Italy*

Nvidia Deep Learning Course — March 2022 – April 2022
Collegio Ghislieri — *Pavia, Italy*

Other Experiences & Volunteering

Treasurer of Collegio Ghislieri Scientific Society — September 2021 – September 2023
Collegio Ghislieri — *Pavia, Italy*

Volunteer at *Mensa del Povero* — September 2022 – July 2024
Convent of Canepanova — *Pavia, Italy*

Available Referees

Prof. C. B. Schönlieb (Phd Advisor)	University of Cambridge	cbs31@cam.ac.uk
Prof. C. De Lellis	IAS, Princeton	camillo.delellis@ias.edu
Prof. U. Stefanelli	University of Vienna	ulisse.stefanelli@univie.ac.at

Cambridge, October 13, 2026



Alex Massucco